Mohammed Sufiyan

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LinkedIn · GitHub · Google Scholar · Portfolio

EDUCATION

Boston University · MS, Applied Data Analytics · GPA 3.9 — **Boston, MA** — *Sep 2025–Dec 2026*

Presidency University · B.Tech, Computer Science & Engineering (Data Science) — **Bengaluru, India** — *Sep 2020–Jul 2024*

EXPERIENCE

Boston University · Graduate Teaching Assistant · MET CS 688 — **Boston, MA** — *Sep 2026–Present*

- Support instruction, student questions, grading, and faculty coordination for a 35-student section.

Boston University · Graduate Research Assistant · CleanCampus — **Boston, MA** — *Jan 2026–Aug 2026*

- Built an IoT monitoring dashboard with Node.js APIs, PostgreSQL, DigitalOcean, and Leaflet to visualize live BU campus heat readings.
- Modeled five PostgreSQL tables and added monitoring filters, trend charts, alerts, and CSV export; presented findings at BU and ACSP 2026.

Boston University Housing · Summer/Fall Operations Assistant · StarRez — **Boston, MA** — *May 2026–Present*

- Built DBMS-backed workflows for room/furniture records, assignments, check-in/out updates, and staff task tracking across BU dorm operations.

Power Tech · Data Engineer — **Bengaluru, India** — *Jun 2024–May 2025*

- Automated 15+ Python data-cleaning workflows, reducing processing time 40%; optimized 20+ SQL queries and built five Power BI dashboards.
- Processed 10GB+ daily with AWS S3/Lambda ETL and Dockerized four apps, cutting deployment time from two hours to under 30 minutes.

SELECTED PROJECTS

Brazilian E-Commerce Data Quality & ETL Validation Framework

Snowflake · dbt · Python/pytest · AWS S3 · PostgreSQL

- Built a synthetic e-commerce warehouse using PostgreSQL source data and seven Parquet datasets uploaded to AWS S3; generated 65.4M records for local-scale validation.
- Developed 10 dbt models, a customer history snapshot, and automated quality coverage; all 46 dbt nodes and 35 dbt data tests passed.
- Validated a 65,443-row Snowflake sample with 59 SQL checks and exact reconciliation across seven tables; all checks passed. Python regression suite: 97 tests passed.

Card Fraud Detection & Alerting Platform · Python, Kafka/Redpanda, PostgreSQL, Redis, FastAPI

- Built an event-driven scoring workflow for simulated transactions, with PostgreSQL audit storage, Redis alert caching, and investigator views in FastAPI/Streamlit.

Pipeline Health Monitor · Python, Pandas, PostgreSQL, FastAPI

- Implemented schema, type, and domain validation for CSV feeds, routing clean and quarantined records with actionable error details.

Formula 1 Pit Stop Analytics · SQL Server, ETL, Star Schema, Python

- Designed normalized and analytical schemas, ETL staging, and ML features for slow pit-stop prediction.

RESEARCH

- **IEEE publication:** Pix2Pix-GAN-Based Lung Segmentation for Chest X-Rays · IEEE, 2025.
- **Ongoing:** LLM-guided policy shaping and reward design in PPO-based reinforcement learning.

TECHNICAL SKILLS

Languages: Python, SQL, R, JavaScript **Data:** dbt, ETL, data modeling, pytest, Pandas, data reconciliation
Databases: PostgreSQL, SQL Server, Snowflake, MySQL, MongoDB, Redis **Cloud & tools:** AWS S3/Lambda, Docker, Git, DigitalOcean
Analytics: Power BI, Tableau, Excel, Leaflet